

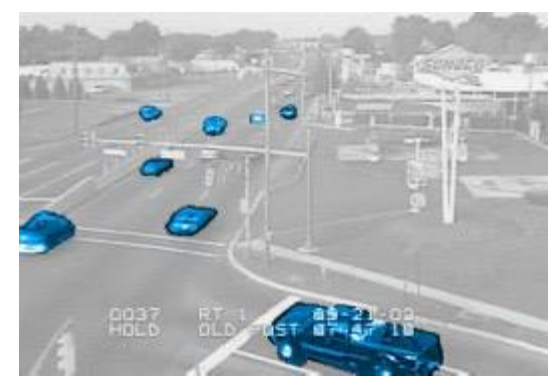
# Networks in SUMO

fundamentals, netedit, netgenerate and  
netconvert

# Elements of a Traffic Simulation

Three elements needed:

- **Network** (roads, intersections, ...) where movements take place
- **Infrastructure** (traffic lights, toll stations, bus stops, ...) and rules to govern traffic behavior
- **Demand** (routes, trips, flows, ...)



# SUMO config file

The **.sumocfg** file specifies all the necessary input files for the simulation + some other configuration parameters.

```
<?xml version="1.0" encoding="UTF-8"?>

<sumoConfiguration xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xsi:noNamespaceSchemaLocation="http://sumo.dlr.de/xsd/sumoConfiguration.xsd">

  <input>
    <net-file value="osm.net.xml.gz"/>
    <route-files value="osm_pt.rou.xml,osm.passenger.trips.xml"/>
    <additional-files value="osm.poly.xml.gz,osm_stops.add.xml,output.add.xml"/>
  </input>

  <output>
    <tripinfo-output value="tripinfos.xml"/>
    <stop-output value="stopinfos.xml"/>
    <statistic-output value="stats.xml"/>
  </output>

  <processing>
    <ignore-route-errors value="true"/>
    <tls.actuated.jam-threshold value="30"/>
  </processing>

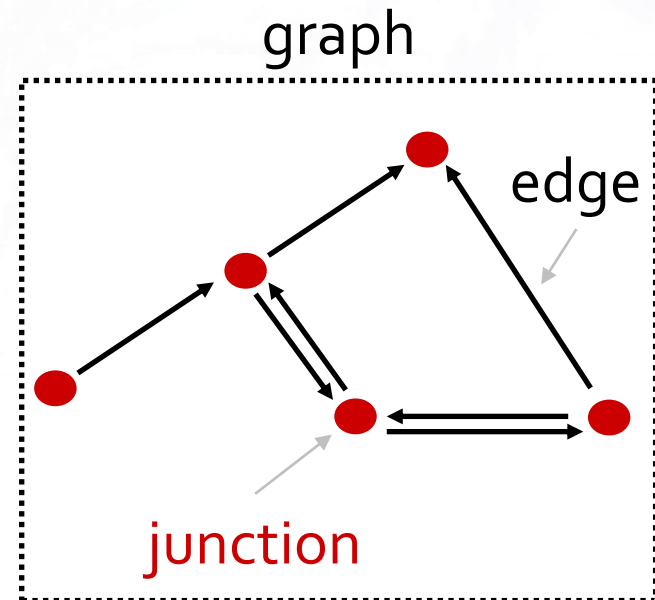
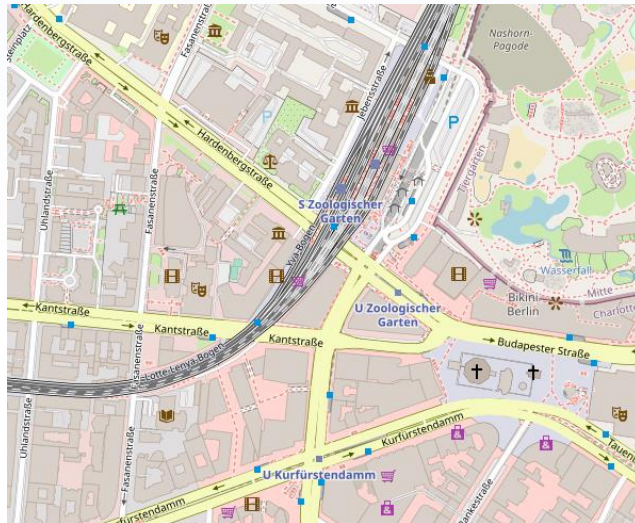
  <routing>
    <device.rerouting.adaptation-steps value="18"/>
    <device.rerouting.adaptation-interval value="10"/>
  </routing>

  <report>
    <verbose value="true"/>
    <duration-log.statistics value="true"/>
    <no-step-log value="true"/>
  </report>

  <gui_only>
    <gui-settings-file value="osm.view.xml"/>
  </gui_only>

</sumoConfiguration>
```

# Basics (I)

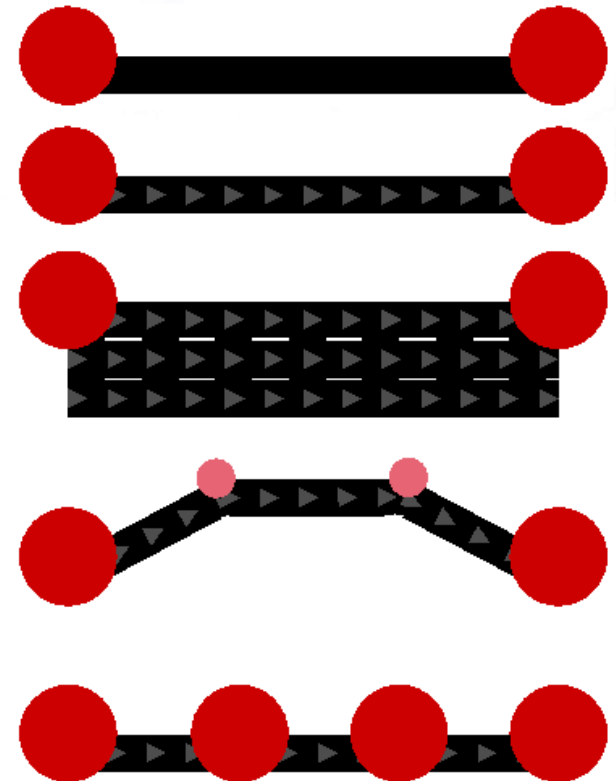


- SUMO road networks are **graphs** of **edges** and **junctions**
- Create SUMO networks with **netedit** (or import using tools)
- They are stored in **.net.xml** files

## Basics (II)

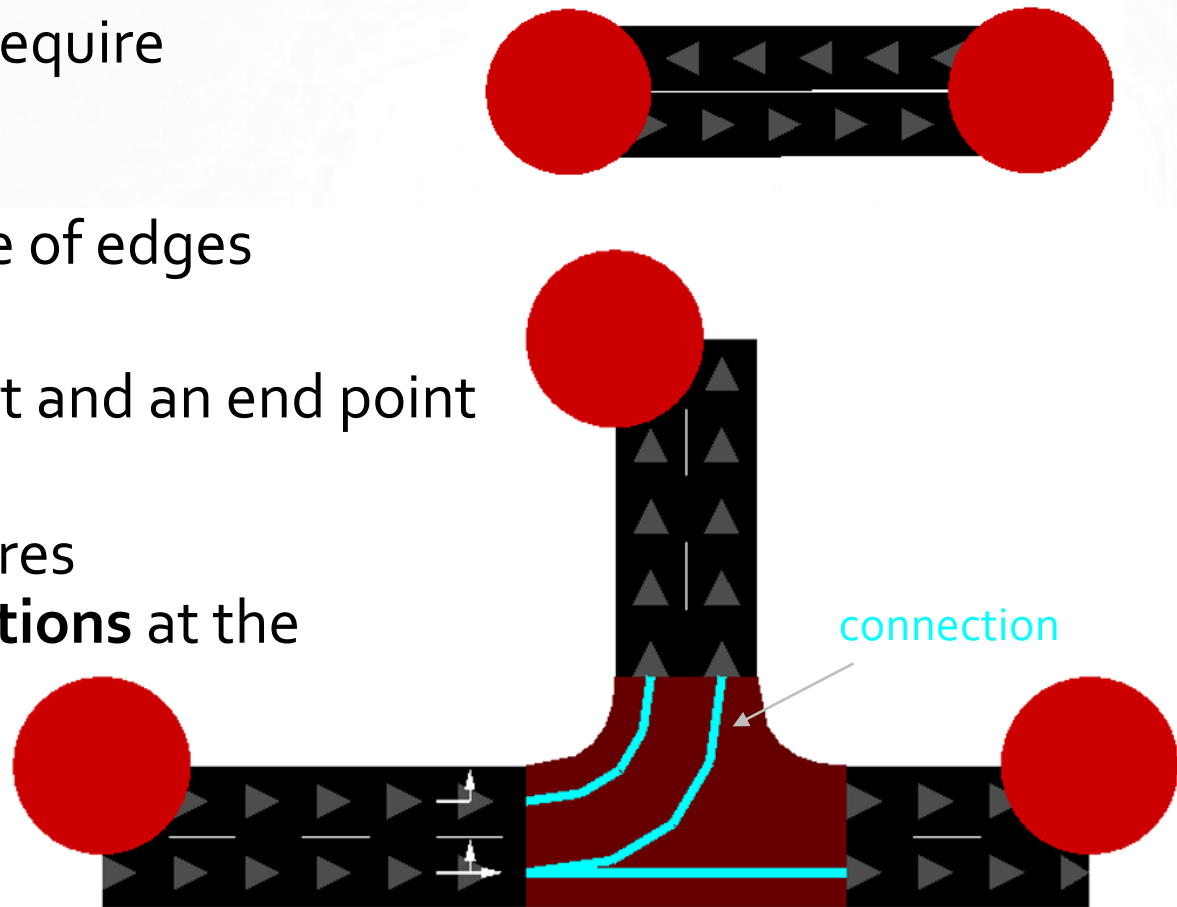
**Edges** are a basic element of the network:

- ... **start/end** at a junction
- ... have **properties** (e.g. id, speed limit)
- ... are **unidirectional**
- ... can have **multiple lanes**
- ... can have **geometry points**
- 3 edges and 4 junctions / nodes



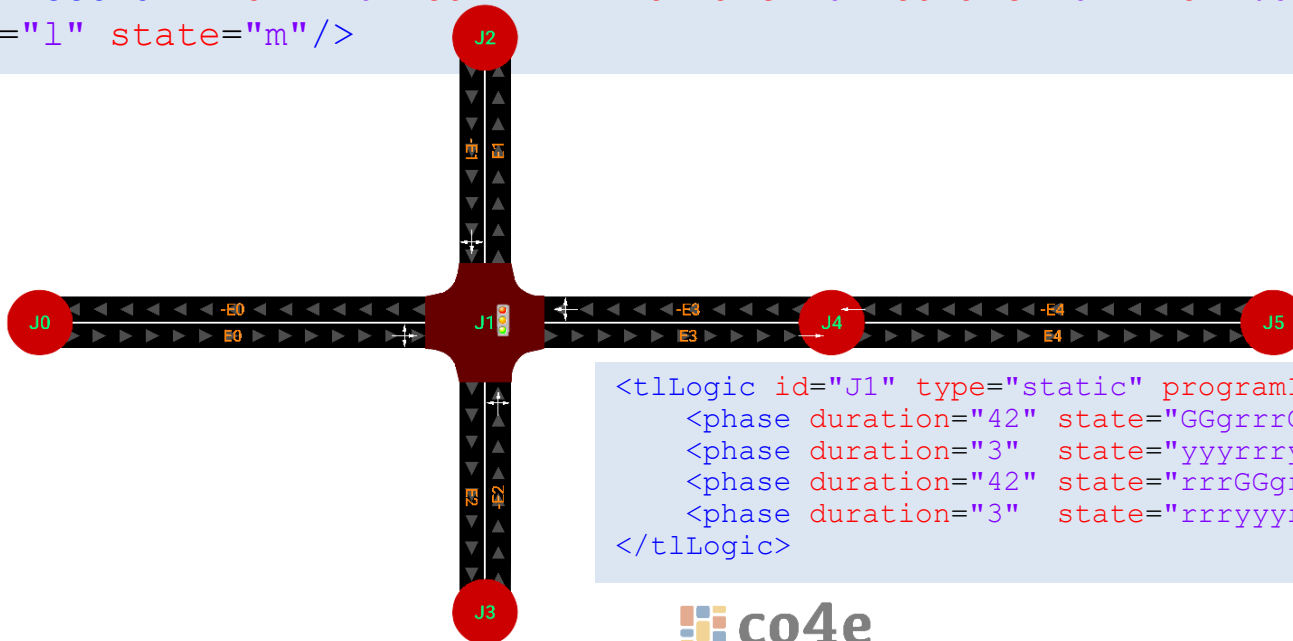
## Basics (III)

- **Bidirectional** roads require (at least) two edges
- A **route** is a sequence of edges
- A **trip** specifies a start and an end point
- Vehicle routing requires **lane-to-lane connections** at the junctions



# Basics (IV)

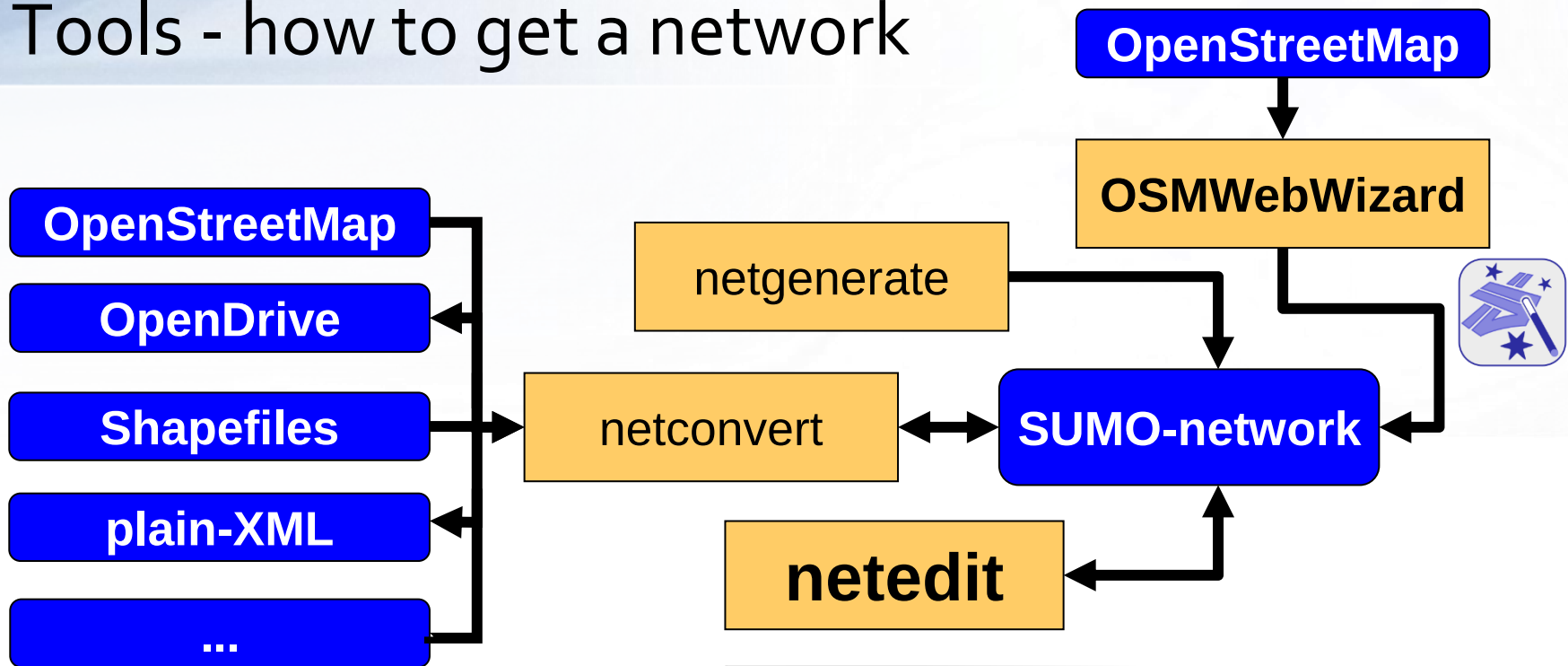
```
<edge id="E3" from="J1" to="J4" priority="-1">
  <lane id="E3_0" index="0" speed="13.89" length="34.60"
    shape="-22.80,-1.40 11.80,-1.40"/>
</edge>
...
<junction id="J2" type="dead_end" x="-30.00" y="34.57" incLanes="E1_0"
intLanes="" shape="-30.00,34.57 -26.80,34.57 -30.00,34.57"/>
...
<connection from="E0" to="E1" fromLane="0" toLane="0" via=":J1_11_0"
dir="l" state="m"/>
```



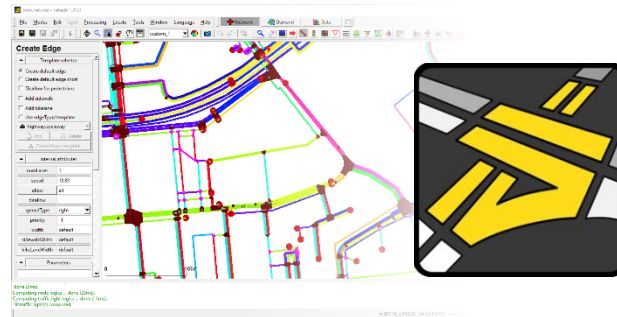
```
<tlLogic id="J1" type="static" programID="0" offset="0">
  <phase duration="42" state="GGrrrrGGrrrr"/>
  <phase duration="3" state="yyyrrrryyyrrr"/>
  <phase duration="42" state="rrrGGrrrrGGg"/>
  <phase duration="3" state="rrrryyyrrrryy"/>
</tlLogic>
```



# Tools - how to get a network



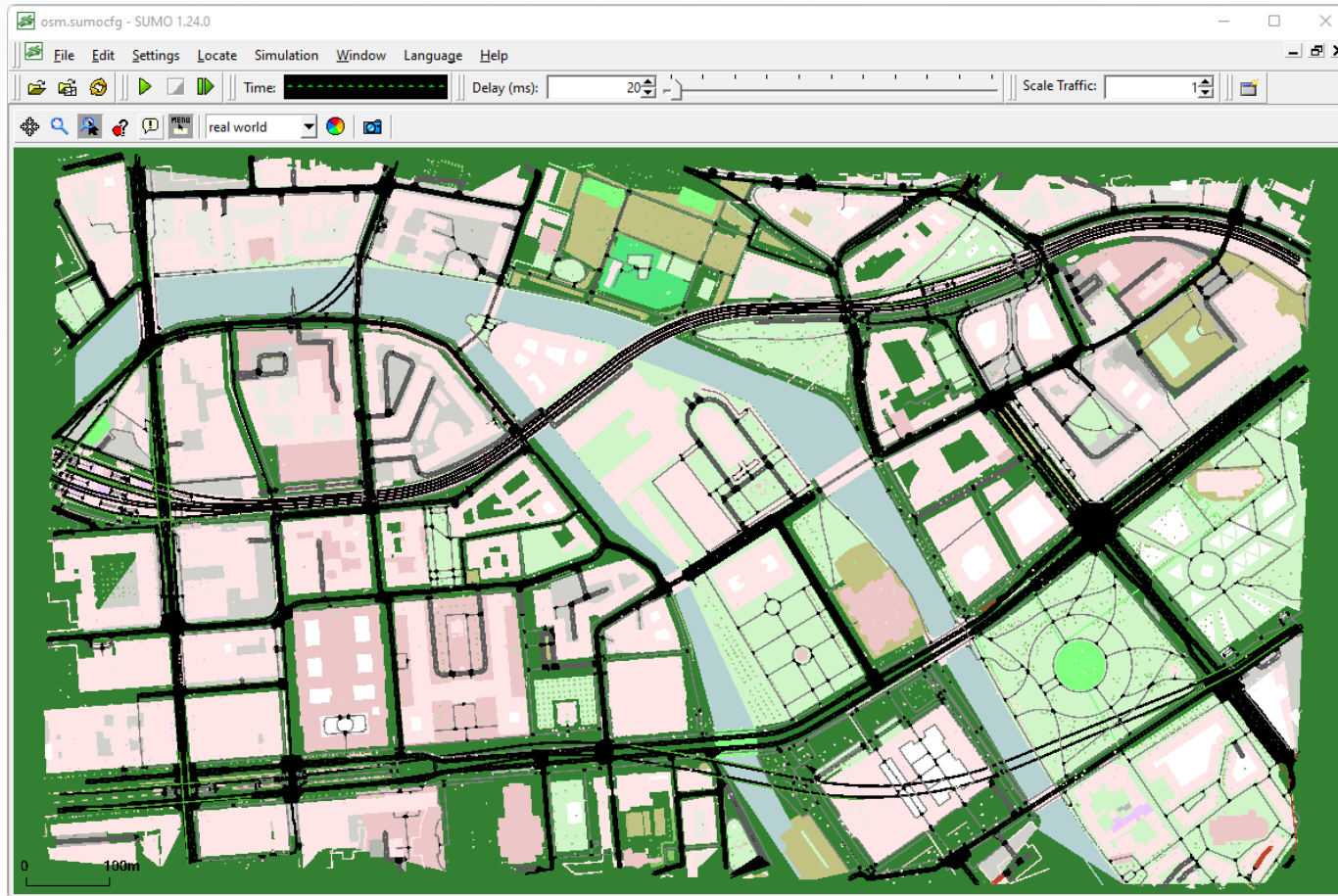
- Import Automation
  - Lane Connections
  - Right-of-Way
  - Traffic light plans





# OSMWebWizard

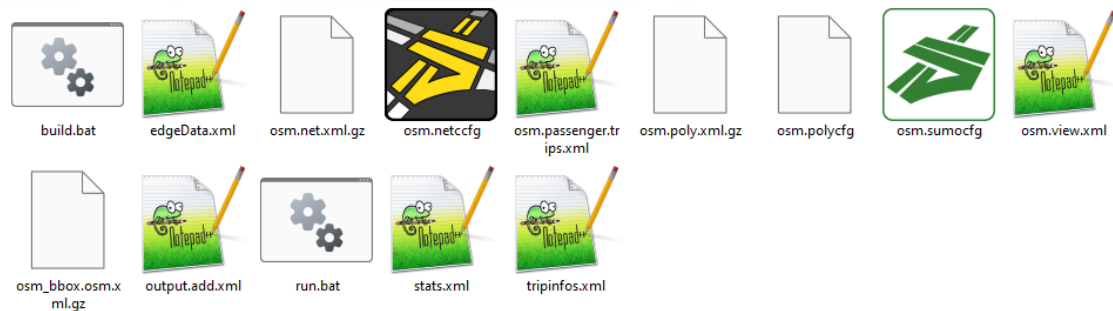
## The fastest way to get started



# OSMWebWizard

## The fastest way to get started

- A new folder (named with the current timestamp) will contain all generated files (e.g. 2025-11-05-09-00-00)

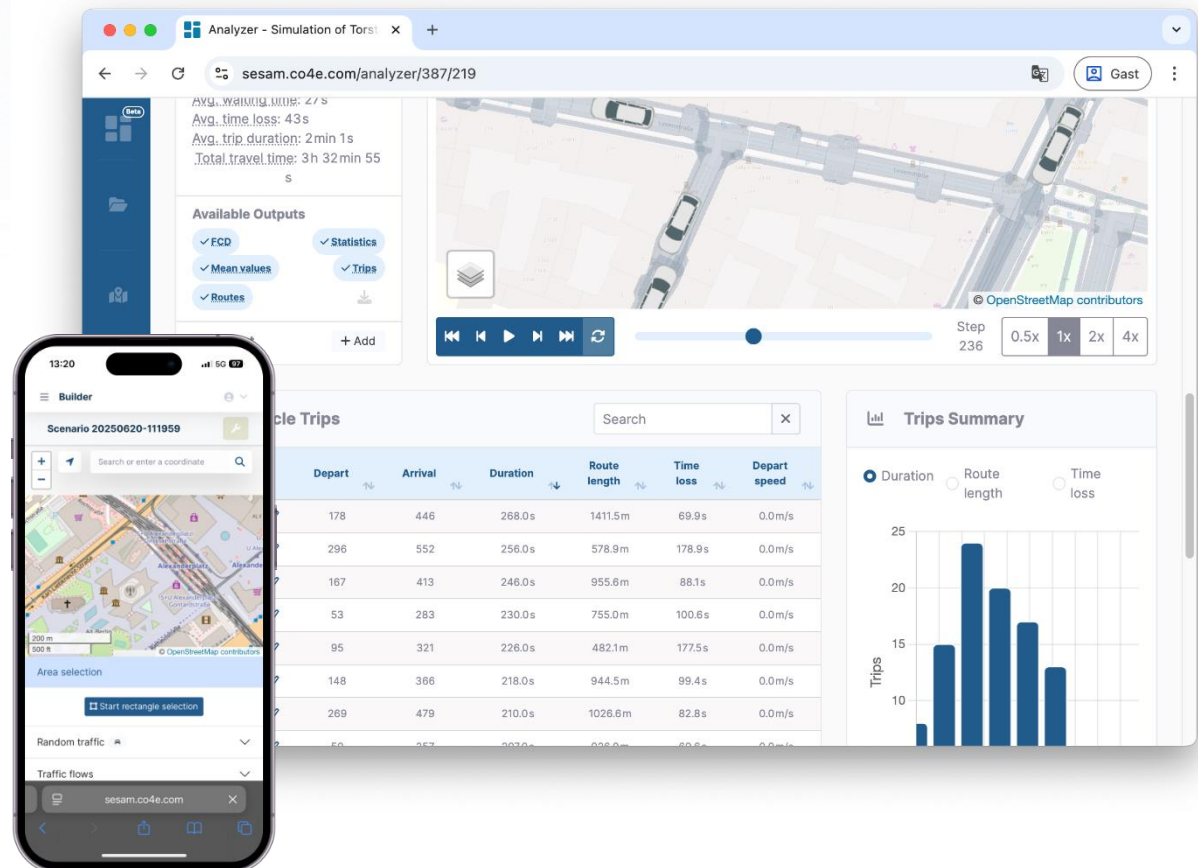


- The generated demand is random (using *randomTrips.py*)
- Always run `build.bat` after modifying the network

# Other quickstart: SESAM

- No installation required
- On the Browser
  - <https://sesam.co4e.com>
- Free

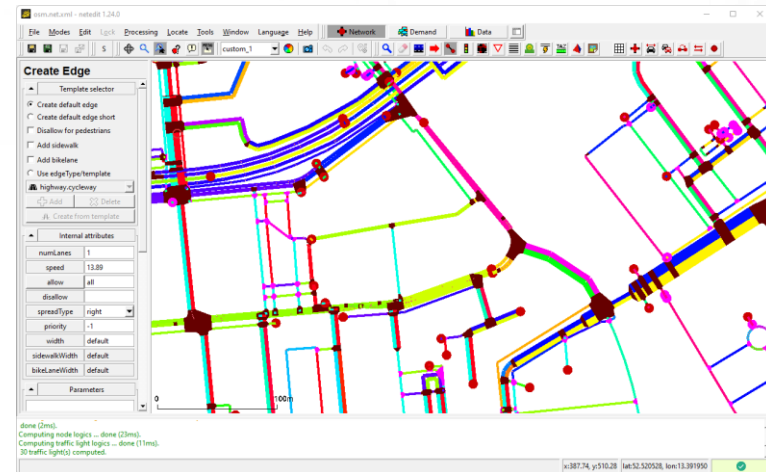
Still in Beta



# Graphical Scenario Editor (netedit)



- Create / edit / view
  - Simulation road/rail networks
  - Infrastructure (traffic lights, stations, parking, ...)
  - Traffic demand
  - Network related Data
- Help: **F1**
- Click: mode-specific action/selection
- Right-click: object context menu
- Pan: click + drag (or use arrows)
- Zoom: right-click + drag (or scroll wheel)
- **F2** / **F3** / **F4** switch between network/demand/data editing
- **F5** re-compute Junctions
- **Ctrl+T** to open *netedit* from *sumo-gui* or *sumo-gui* from *netedit*



supermodes

specific modes

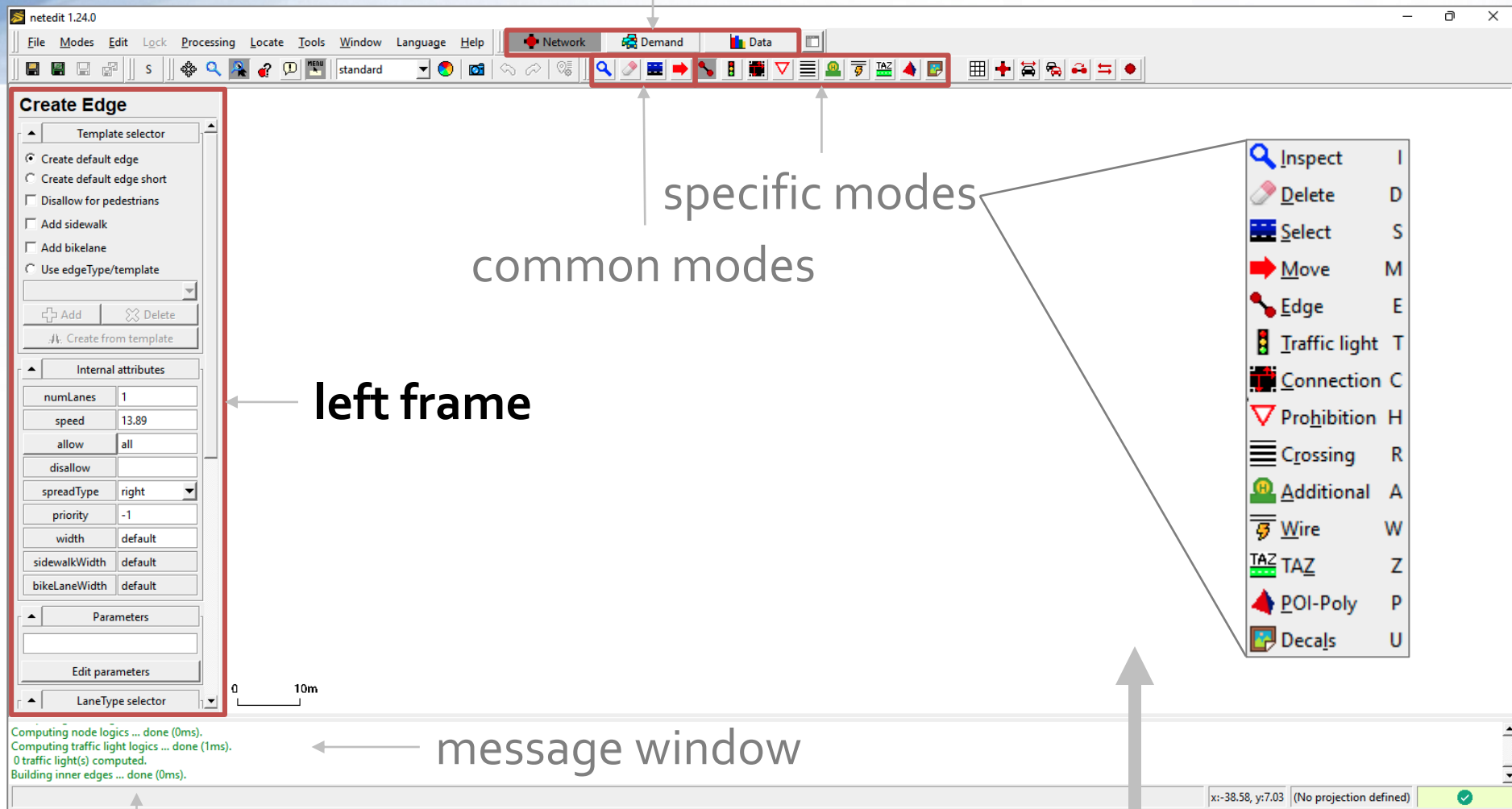
common modes

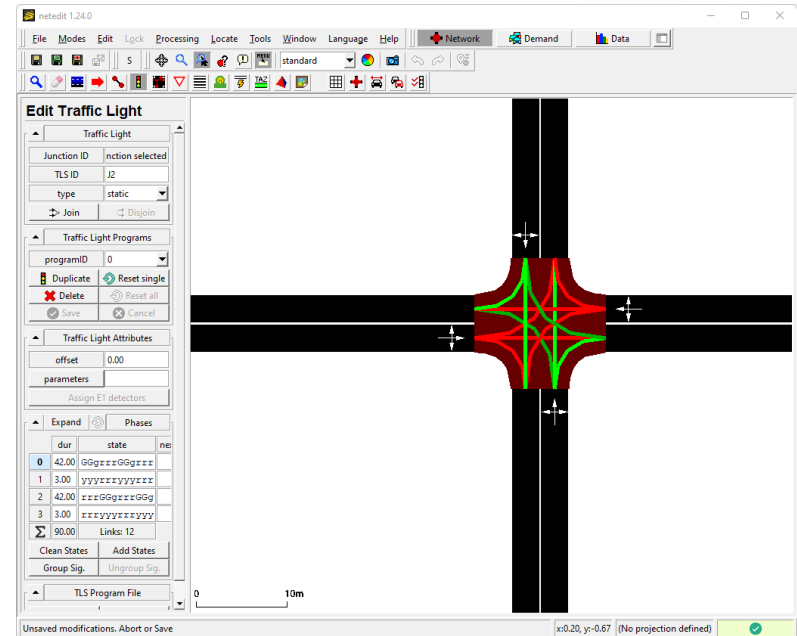
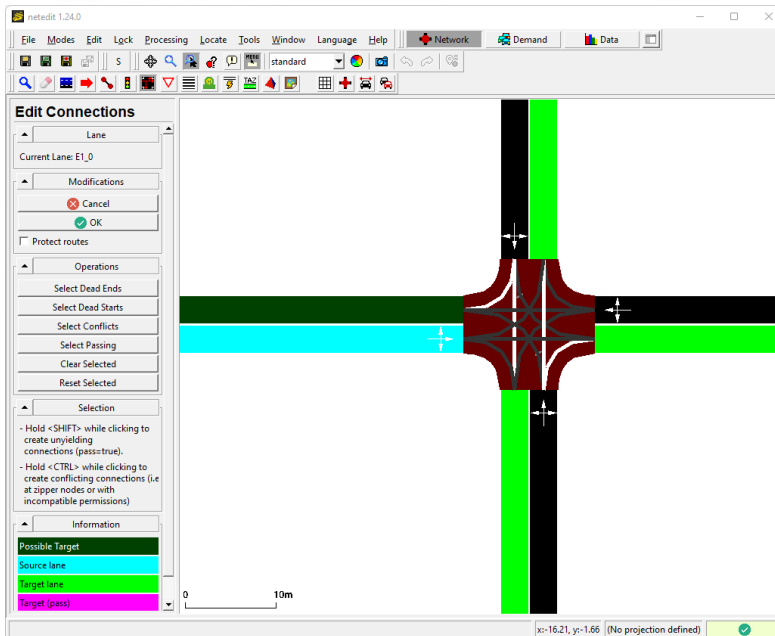
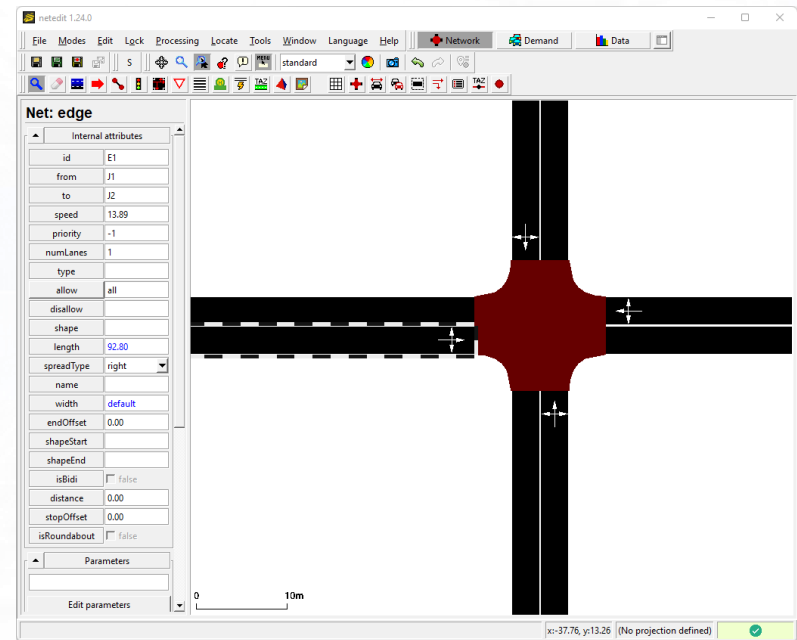
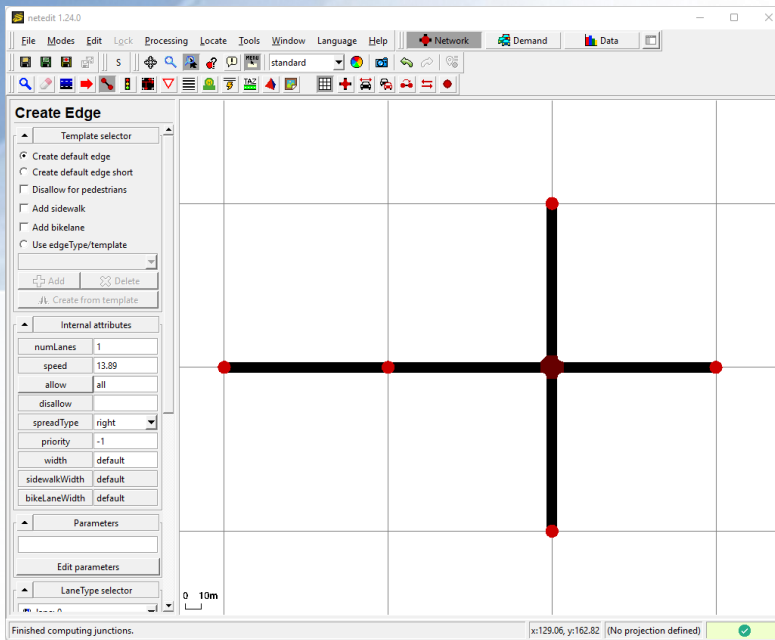
left frame

message window

status line

view

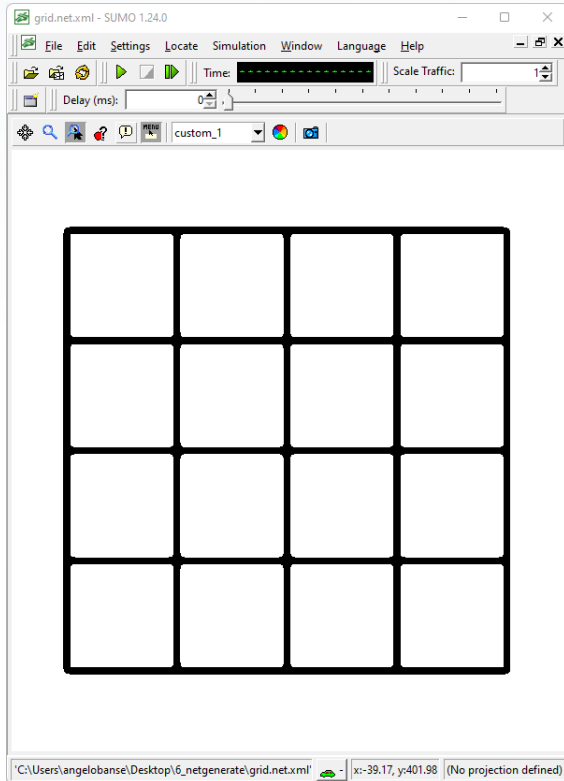




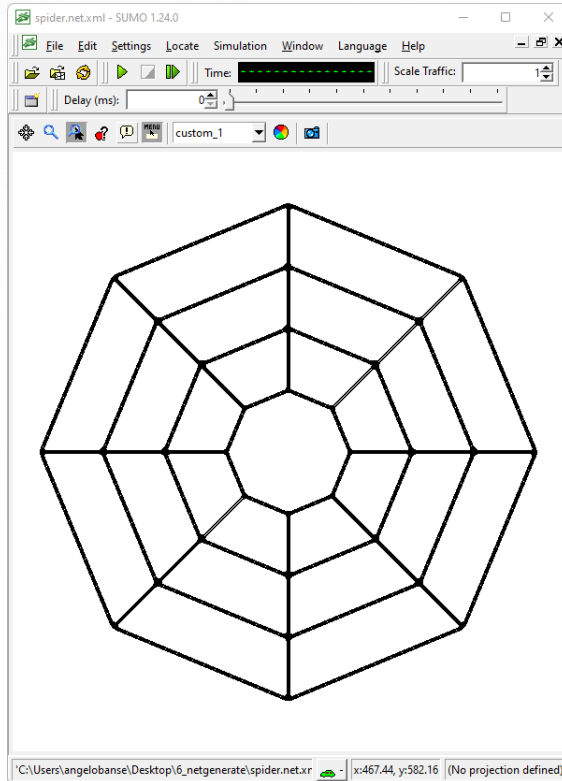
# DEMO



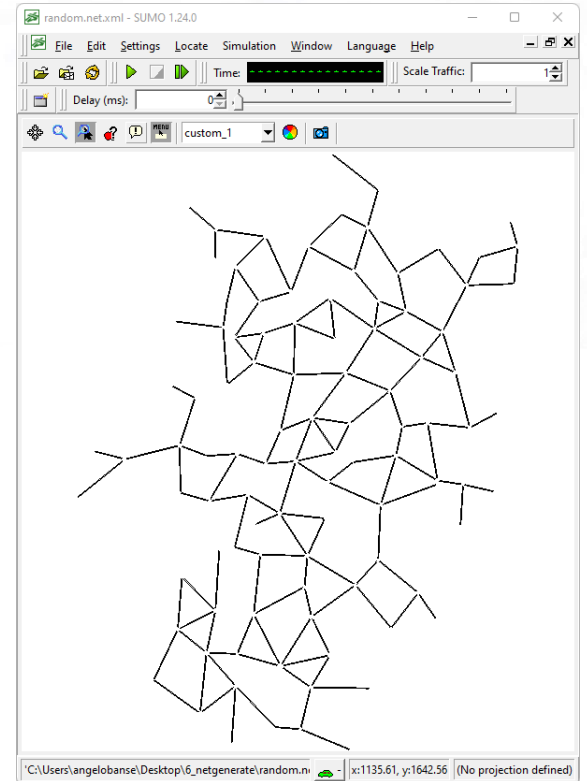
# netgenerate – Generate Abstract Road Networks



Grid



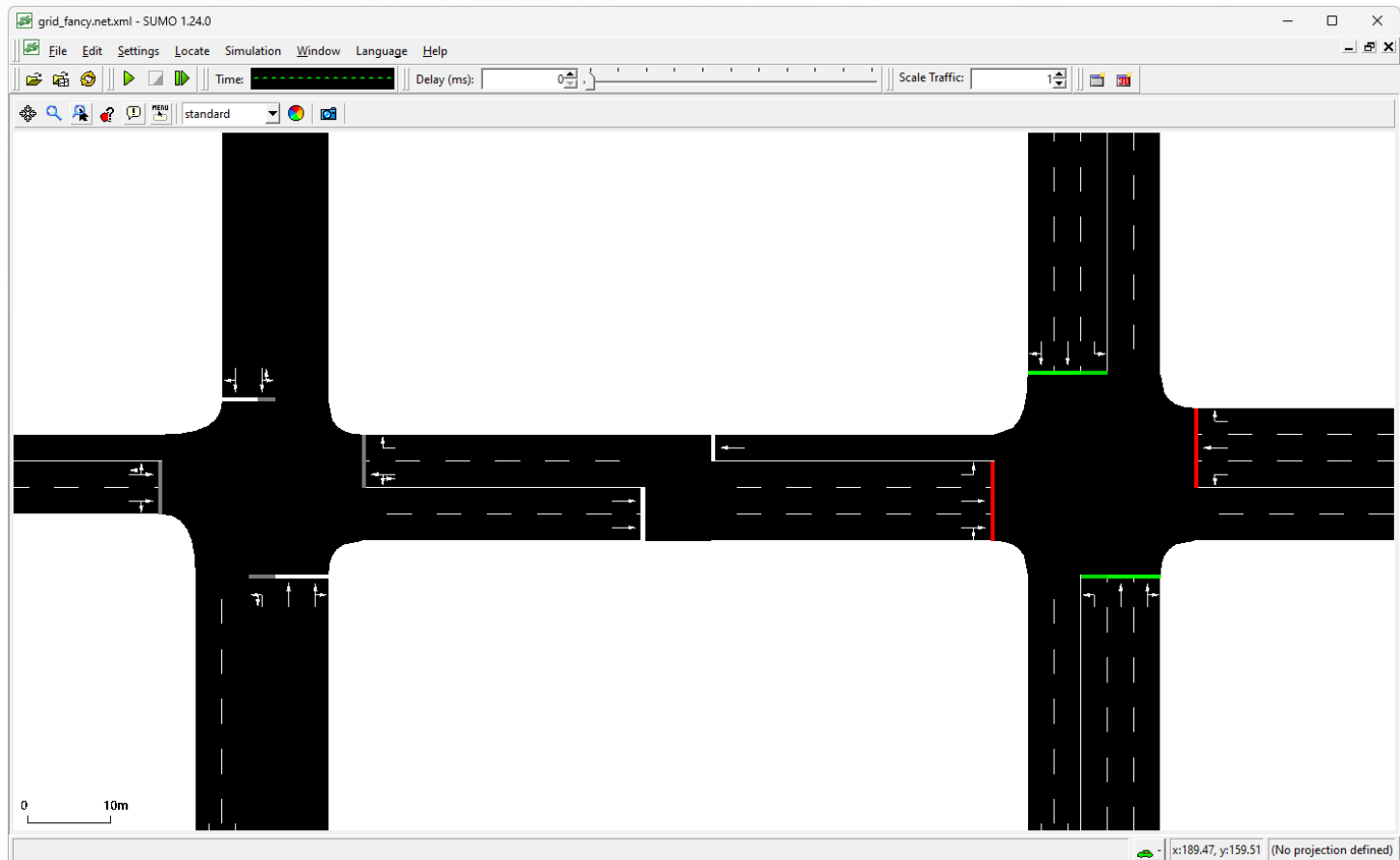
Spider



Random

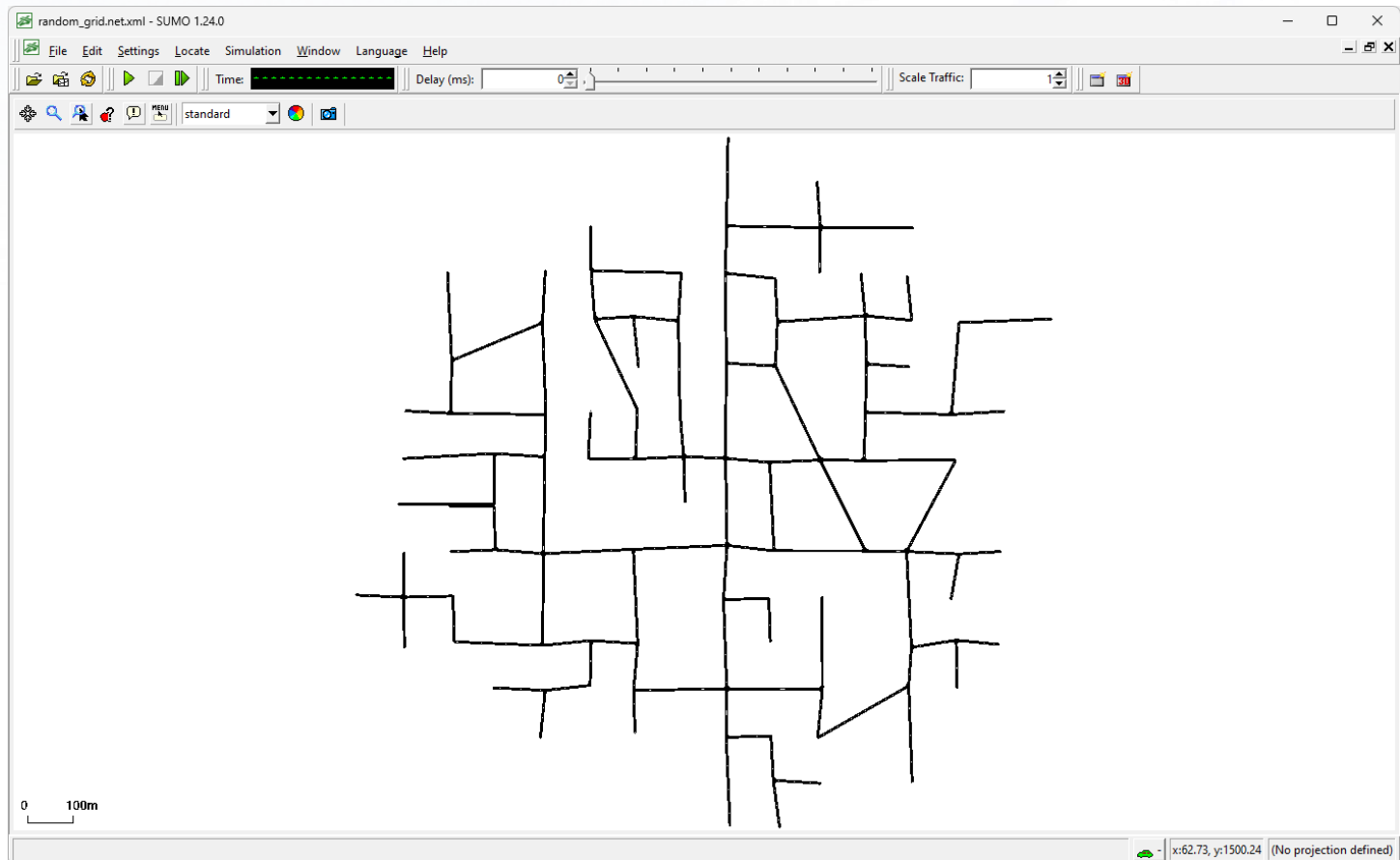
# netgenerate

- options: **--turn-lanes** **--random-lanenumber**



# netgenerate

- options: `--grid` `--perturb-x` `--perturb-y` `--perturb-z`



# netconvert: Supported Network Formats

- OpenStreetMap (OSM)
- Shapefile (ArcGIS)
- OpenDrive
- VISUM/VISSIM
- MATSIM
- Simple xml format (.nod.xml, .edg.xml)

```
<node id="a" x="0.0" y="0.0" type="traffic_light"/>
```

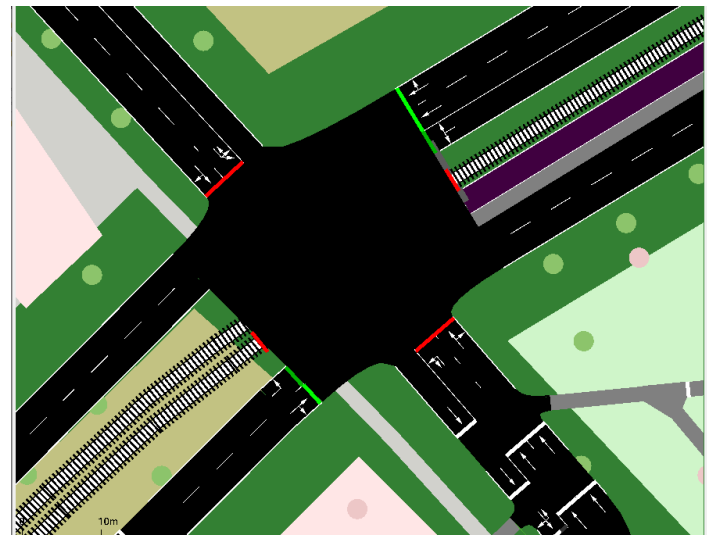
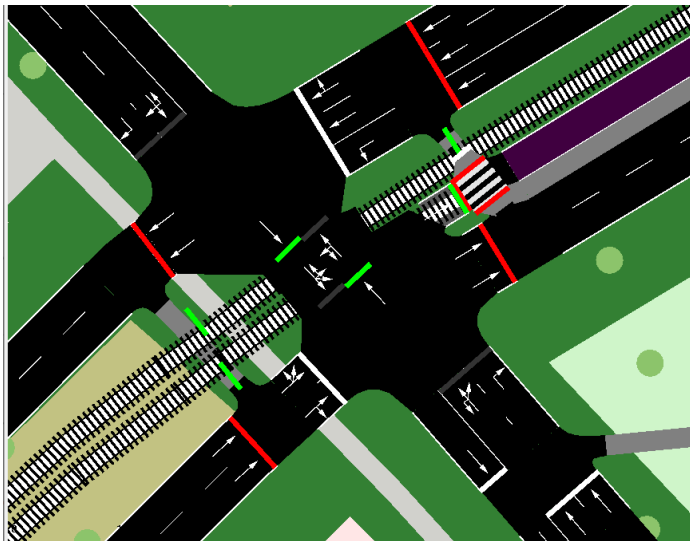
```
<edge id="ab" from="a" to="b" priority="3" numLanes="2" speed="13.89"/>
```

- .net.xml (SUMO network format) - automated reprocessing / conversion

# Manual correction of OSM Networks

- Complex junction clusters must be joined manually
- Traffic light plans lack coordination (or are simply bad for complex junction)
- Missing / superfluous traffic lights
- Wrong lane count (missing turn lanes)
- Wrong lane-to-lane connection

Import using the [recommended options](#)



# Q & A